Group-3 CoMeg Assignment-3

Members:

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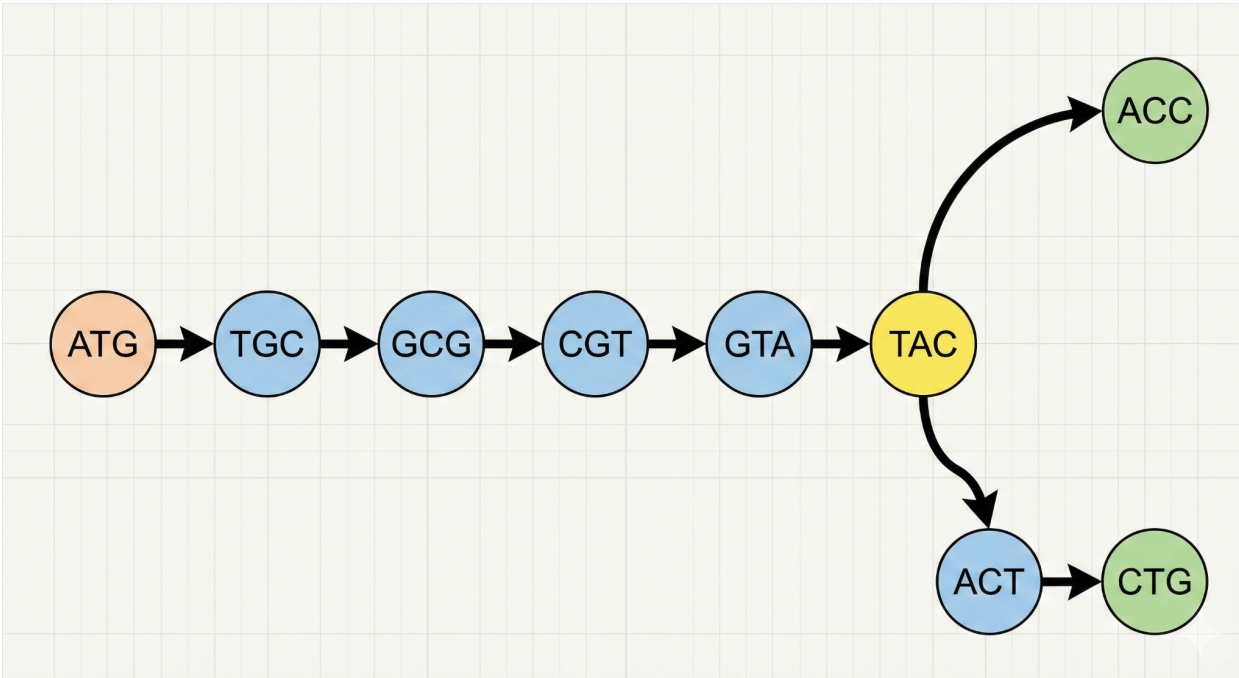
Section 1 – De Bruijn graph (k = 4)

Reads (6 bp):
R1: ATGCGT
R2: TGCGTA
R3: CGTACC
R4: GTACTG

Q1. All k-mers and De Bruijn graph
k-mers (length 4):

- From R1 (ATGCGT): ATGC, TGCG, GCGT
- From R2 (TGCGTA): TGCG, GCGT, CGTA
- From R3 (CGTACC): CGTA, GTAC, TACC
- From R4 (GTACTG): GTAC, TACT, ACTG

Unique k-mers:
ATGC, TGCG, GCGT, CGTA, GTAC, TACC, TACT, ACTG



Nodes are (k–1)-mers (length 3). For each k-mer, edge = prefix → suffix:

- ATGC: ATG → TGC
- TGCG: TGC → GCG
- GCGT: GCG → CGT
- CGTA: CGT → GTA
- GTAC: GTA → TAC
- TACC: TAC → ACC
- TACT: TAC → ACT
- ACTG: ACT → CTG

Adjacency list representation:

- ATG → TGC
- TGC → GCG
- GCG → CGT
- CGT → GTA
- GTA → TAC
- TAC → ACC, ACT
- ACT → CTG
- ACC → (no outgoing edge)
- CTG → (no outgoing edge)

Q2. Likely contig(s)

Following the graph, there is a branch at node TAC, so we get two possible paths:

1. ATG → TGC → GCG → CGT → GTA → TAC → ACC

- Contig: ATGCGTACC

2. ATG → TGC → GCG → CGT → GTA → TAC → ACT → CTG
   - Contig: ATGCGTACTG

So, the graph suggests two contigs:
ATGCGTACC and ATGCGTACTG.

Q3. Why reads give better assemblies than only k-mers?
Reads preserve the original order of overlapping k-mers and their co-occurrence on the same fragment. This extra context (plus read coverage information) helps distinguish true paths from spurious ones, resolve some repeats, and handle sequencing errors better. Using only k-mer counts loses this linkage and makes the graph more ambiguous.

Q4. One challenge from repeats in De Bruijn graphs
Repeats of length ≥ k collapse into shared nodes/edges, merging different genomic regions into a single path. This creates branches or bubbles that make it unclear how to traverse the graph, often leading to chimeric contigs or fragmented assemblies because the assembler can't uniquely resolve which path corresponds to which genomic locus.

Section 2 – Contig Binning (GC% & coverage)

Table (from assignment):
CoMeG_Assignment3

Contig	GC%	Cov-S1	Cov-S2
C1	36	12	11
C2	35	13	12
C3	37	11	10
C4	52	40	8
C5	53	42	9
C6	51	39	7
C7	44	18	17
C8	43	19	16

Q1. Cluster into three genome bins
A natural clustering:

- Bin A: C1, C2, C3
- Bin B: C4, C5, C6
- Bin C: C7, C8

Q2. Justification (GC% + coverage)
GC% patterns

- Bin A (C1–C3): Low GC ~35–37%.
- Bin B (C4–C6): High GC ~51–53%.
- Bin C (C7–C8): Intermediate GC ~43–44%.

Coverage patterns

- Bin A: Cov-S1 ~11–13 and Cov-S2 ~10–12; similar abundance in both samples.
- Bin B: High Cov-S1 (~39–42) but low Cov-S2 (~7–9); strongly enriched in S1.
- Bin C: Medium coverage in both samples (S1: 18–19, S2: 16–17); similar S1/S2 pattern distinct from other bins.

Q3. Conceptual
Why multi-sample coverage helps vs single-sample
With only one sample, contigs from different genomes that have similar coverage and GC% can look indistinguishable. Using multiple samples adds a coverage profile for each contig (e.g., high in S1, low in S2, etc.). Contigs from the same genome tend to share similar patterns across samples, so multi-sample coverage lets us cluster contigs more reliably and separate strains with similar composition but different abundances.

Limitation of GC% alone

GC% is a coarse compositional feature. Unrelated species can have similar GC%, and within a single genome, local GC% can vary (islands, plasmids, phages). Horizontal gene transfer or mobile elements may have very different GC% from the host, causing them to be mis-binned. Thus, GC% alone can group unrelated contigs together or split a single genome into multiple bins.

Section 3 – Machine Learning Models

Q1:
We stacked the four species abundance matrices (LloydPriceJ_2019, SongJ_2023, WallenZ_2020, YachidaS_2019) by rows and outer-joined all species columns, resulting in a 2566×2218 abundance matrix. We then concatenated the four metadata tables (2566×3) and merged them with the abundance matrix by sample ID, yielding a final combined dataset of 2566 samples × 2221 columns (2218 species + 3 metadata columns: study_name, age, disease)

```
Combined species shape: (2566, 2218)
Combined meta shape: (2566, 3)
Final merged dataset shape: (2566, 2221)
Disease label counts:
 disease
1    1386
0    1180
Name: count, dtype: int64
Train shape: (2052, 2218) Test shape: (514, 2218)
```

Q2:
a:
Disease Classification (Binary: Control vs Disease)

Model	Accuracy	AUC
Random Forest	0.6907	0.7826
XGBoost	0.6926	0.7884

Interpretation:
For disease classification, both RF and XGBoost achieved reasonably high accuracy and AUC.
In our run, XGBoost achieved the higher AUC (0.7884), performing slightly better than RF, likely due to its ability to capture nonlinear feature interactions among microbial species.

B:
Age Prediction (Regression)

Model	MAE	Correlation
Random Forest	**8.8508**	**0.7046**
XGBoost	**8.6967**	**0.7176**

Interpretation:
For age prediction, both models showed moderate correlation between actual and predicted age.
In our run, XGBoost achieved lower MAE and higher correlation, indicating better regression performance and stronger predictive consistency compared to Random Forest.

Across both disease classification and age regression tasks, XGBoost consistently performed slightly better than Random Forest. For disease detection, XGBoost had a higher AUC (0.7884 vs 0.7826), suggesting superior ability to capture nonlinear species–disease relationships. For age prediction, XGBoost achieved lower MAE (8.70 vs 8.85) and higher correlation (0.718 vs 0.705), indicating stronger predictive accuracy. Overall, XGBoost provided the best performance in our dataset, though both models produced biologically meaningful results.

Q3:

Feature importance for Random Forest was computed using Gini impurity (model.feature_importances_). Feature importance for XGBoost was obtained using the built-in importance scores based on gain, which measures how much each feature improves the model's decision splits.

RF -Top 10 species for disease (plot 1, top)
The Random Forest model ranks *Anaerostipes_hadrus*, *Eubacterium_hallii*, *Eubacterium_ventriosum*, *Fusicatenibacter_saccharivorans*, *Blautia_luti*, *Faecalibacterium_prausnitzii*, *Coprococcus_eutactus*, *Bacteroides_thetaiotaomicron*, *Streptococcus_thermophilus*, and *Dorea_longicatena* as the most important features for disease classification.

XGB- Top 10 species for disease (plot 1, bottom)
XGBoost highlights *Bacteroides_* (top-ranked), *Eubacterium_desmolans*, *Lactobacillus_johnsonii*, *Blautia_luti*, *Eubacterium_dolichum*, *Pseudomonas_aeruginosa*, *Eubacterium_ventriosum*, *Oligella_urethralis* (and a couple of other species) as important for classifying disease.

RF- Top 10 species for age (plot 2, top)
For age prediction RF places *Pseudomonas_aeruginosa* highest, followed by *Streptococcus_salivarius*, *Firmicutes_bacterium_CAG_83*, *Shigella_flexneri*, *Mogibacterium_diversum*, *Streptococcus_parasanguinis*, *Roseburia_intestinalis*, *Faecalibacterium_prausnitzii*, *Roseburia_faecis*, and *Fusicatenibacter_saccharivorans*.

XGB- Top 10 species for age (plot 2, bottom)
XGBoost for age similarly ranks *Pseudomonas_aeruginosa* first, then *Shigella_flexneri*, *Streptococcus_salivarius*, *Mogibacterium_diversum*, *Firmicutes_bacterium_CAG_83*, (a *Clostridium*-related taxon), *Shigella_sonnei*, *Streptococcus_gordonii*, *Streptococcus_parasanguinis*, and *Streptococcus_oralis*.

Cross-model / cross-task patterns (important repeats)

Species that appear repeatedly across models/tasks:

Pseudomonas_aeruginosa - top for age in both RF and XGBoost.

Shigella spp. (e.g., *S. flexneri*, *S. sonnei*) - repeatedly important for age (both models).

Streptococcus spp. (multiple species) - prominent in age prediction (especially XGB).

Faecalibacterium_prausnitzii, *Blautia_luti*, *Eubacterium_ventriosum*, *Fusicatenibacter_saccharivorans* - show up in disease and/or age lists (RF and XGB), indicating consistent signal across tasks.

- **Model agreement:** XGBoost and Random Forest share several top features for each task (e.g., *Blautia_luti* and *Eubacterium_ventriosum* for disease; *Pseudomonas_aeruginosa*, *Shigella* and *Streptococcus* taxa for age). This concordance increases confidence these taxa carry real predictive information in this combined dataset.

<u>## Section 4 – Leave-One-Study-Out (LOSO)</u>

Q1:

Held-out study	RF Accuracy	RF AUC	XGB Accuracy	XGB AUC
LloydPriceJ_2019	0.4267	0.5038	0.3978	0.5254

SongJ_2023	0.4398	0.4531	0.5746	0.4519
WallenZ_2020	0.4075	0.4957	0.3813	0.4092
YachidaS_2019	0.5642	0.4734	0.5417	0.5018

Interpretation:

For LOSO disease classification, predictive performance dropped noticeably compared to the 80/20 random split, highlighting the challenge of cross-cohort generalization.

RF and XGB both showed low AUCs, typically around 0.45–0.52, indicating limited ability to transfer learned patterns from one cohort to another.
XGBoost provided slightly higher AUC on LloydPriceJ_2019, while Random Forest performed slightly better on WallenZ_2020.
Overall, no model consistently dominated across all held-out studies.

This demonstrates that disease-related signatures in the microbiome are strongly cohort-specific, and models may struggle to generalize across independently collected datasets.

q2:

Held-out study	RF MAE	RF Corr	XGB MAE	XGB Corr
LloydPriceJ_2019	22.22	0.1216	21.42	0.1071
SongJ_2023	21.47	0.0792	19.24	0.0662
WallenZ_2020	24.07	0.0249	24.35	0.0952
YachidaS_2019	22.42	0.2152	22.48	0.2291

Interpretation:

For LOSO age prediction, correlations between actual and predicted age on held-out studies were generally much lower than in the random 80/20 split, reflecting limited cross-study transferability.

Correlations remained weak (typically 0.02–0.23) across all cases, showing that age-related microbiome signatures do not generalize well across cohorts.

XGBoost achieved the highest correlations on `YachidaS_2019 (0.229) and WallenZ_2020 (0.095).
Random Forest showed the highest correlation on LloydPriceJ_2019 (0.122).
MAE remained high (≈19–24 years) across all folds for both models.

Overall, neither RF nor XGB generalised strongly across studies, but XGBoost did slightly better in most folds in terms of correlation, indicating somewhat more robust age-related signal extraction - though still weak and highly dataset dependent.